

JACOB COOK

Detection & Response Engineer | Cloud-First SIEM · EDR · Threat Hunting | Sigma · Python · MITRE ATT&CK · AI-Augmented

Sacramento, CA · 916 709 1794 · jacobcookofficial@gmail.com
[linkedin.com/in/jcook-dev](https://www.linkedin.com/in/jcook-dev) · github.com/jacobdcook · jacobdcook.com

Detection-focused engineer with hands-on experience building alerts, automation workflows, and logging pipelines from scratch in Wazuh, authoring Sigma rules mapped to MITRE ATT&CK, and reducing false positives to zero in simulated triage. Strong on attacker lifecycle, intrusion detection, scripting (Python, Bash, FastAPI), and AI-augmented investigation workflow design. Comfortable consuming threat intel and turning it into detection content. CompTIA-certified across four disciplines (Security+, CySA+, PenTest+, SecurityX); BS Computer Science; MS Cybersecurity in progress.

TECHNICAL SKILLS

Detection & Response: Alert Engineering, SIEM Query Authoring (Sigma, KQL familiar, SPL familiar), MITRE ATT&CK, Attacker Lifecycle, Threat Hunting, Threat Intel Consumption, Logging Pipeline / Source Onboarding, Incident Response, On-call Discipline
Tooling & Telemetry: Wazuh SIEM (build & admin), EDR concepts, Network & Firewall Logs, Wireshark, OSINT, AI-Augmented Investigation

Cloud & Identity: Azure AD / Entra ID, Okta, AWS labs, SSO / RBAC, Terraform, Linux / macOS / Windows internals (familiar)

Development: Python, Bash, SQL, FastAPI, React, Git

TECHNICAL PROJECTS

Stryker / Intune Detection Pack: Detection-as-Code 2026

- Authored 6 Sigma-format detection rules for Intune MDM identity abuse mapped to MITRE ATT&CK, with enrichment logic that eliminated 100% of false positives in a 100-event simulation (80 benign suppressed, 20 malicious caught).
- Documented 3 benign scenarios per rule (18 total) covering VPN logins, scheduled maintenance, and PIM activations so analysts can hunt and respond confidently without escalating known-good activity.
- Verification: 43 passing pytest cases in repo (tests/test_rules.py), reproducible against synthetic data.

Blue Team SOC Monitoring Lab: Wazuh SIEM (End-to-End Build) 2026

- Built a full Wazuh SIEM stack from scratch — log ingestion, normalization, alert rules — and validated 12 detection patterns against 242 simulated auth-style events with 100% precision after tuning.
- Reduced false-positive alert volume by 100% by tuning out 79 benign-only matches without dropping a single true positive — the same logging-pipeline ownership and detection-coverage discipline cloud-first SOCs require.
- Verification: 19 passing pytest cases in repo (tests/test_detection_accuracy.py).

SOAR-lite: Incident Response Orchestrator 2026

- Built internal tooling to expand and automate detection & response capabilities: webhook ingestion from Wazuh, automatic playbook routing, IP blocking, Slack notification, and incident documentation with zero analyst intervention.
- Eliminated four manual triage steps and reduced triage time by over 60% in lab testing — the same automation-workflow pattern modern detection teams ship internally.
- Verification: 10 passing pytest cases in repo (tests/test_orchestrator.py). Tests run against the real FastAPI app, not mocked endpoints.

Phishing Analysis Lab 2026

- Triaged 5 real-world phishing samples end-to-end: header inspection, SPF/DKIM/DMARC validation, URL defanging, VirusTotal enrichment, IOC extraction, attacker infrastructure mapping — 100% accurate verdicts against ground truth.
- Translated triage findings into IOC-driven detection improvements, the same pattern hunters use to convert investigation outputs into new alert content.

G3-GPT: Industry-Sponsored Enterprise AI Platform (Team of 6) 2024

- Engineered RBAC and Azure AD SSO controls on a production AI platform that enabled executive-level adoption with zero critical security findings at delivery — direct cloud-first identity engineering experience.

PROFESSIONAL EXPERIENCE

Target Distribution Center | Operations Engineer: Systems Monitoring & Incident Response 2019 - 2025

- Maintained 99.5% system uptime by triaging telemetry alerts on rotating shifts — the same continuous-monitoring and on-call discipline detection-and-response engineering operates on.
- Resolved 200+ anomalies through structured root cause analysis, producing tracked documentation that reduced repeat incidents by an estimated 30%.
- Reduced mean time to recovery by an estimated 20% through cross-team coordination, escalation, and post-incident reporting that mirrors detection-and-response incident workflows.

EDUCATION & CERTIFICATIONS

WGU | MS, Cybersecurity (In Progress)

Expected Oct 2026 · GPA 3.50

CSUS | BS, Computer Science

Jan 2025 · GPA 3.50

CompTIA: Security+ · CySA+ · PenTest+ · SecurityX · CSIE (stackable)